

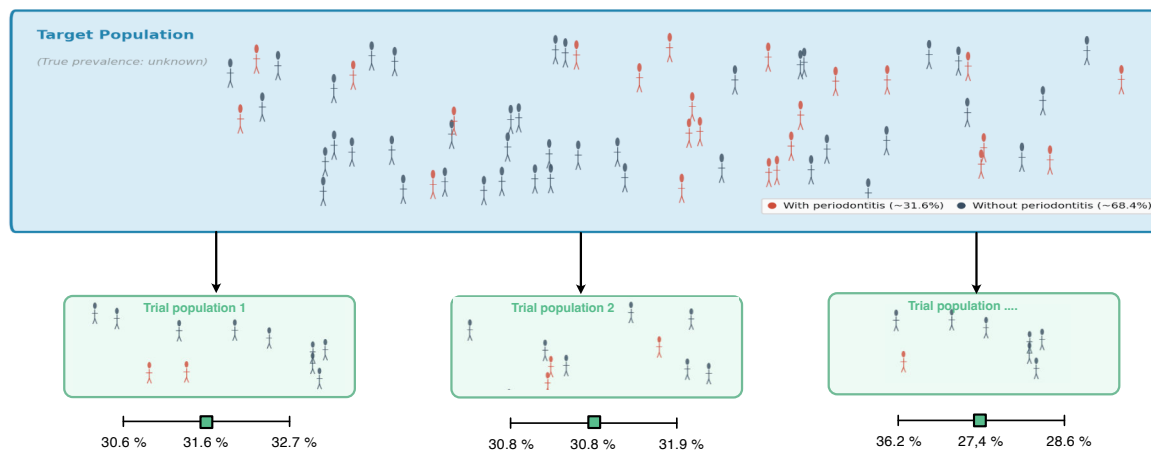
WHAT IS A CONFIDENCE INTERVAL?

A confidence interval gives a reasonable range for the true value in the wider (target) population, based on the sample data (study population).

Confidence Intervals

How confidence intervals arise from repeated sampling - the frequentist definition

The point estimate is the single best estimate. The confidence interval shows how much uncertainty surrounds that estimate.



Simulated 95% confidence intervals - 10 Hypothetical study repetitions

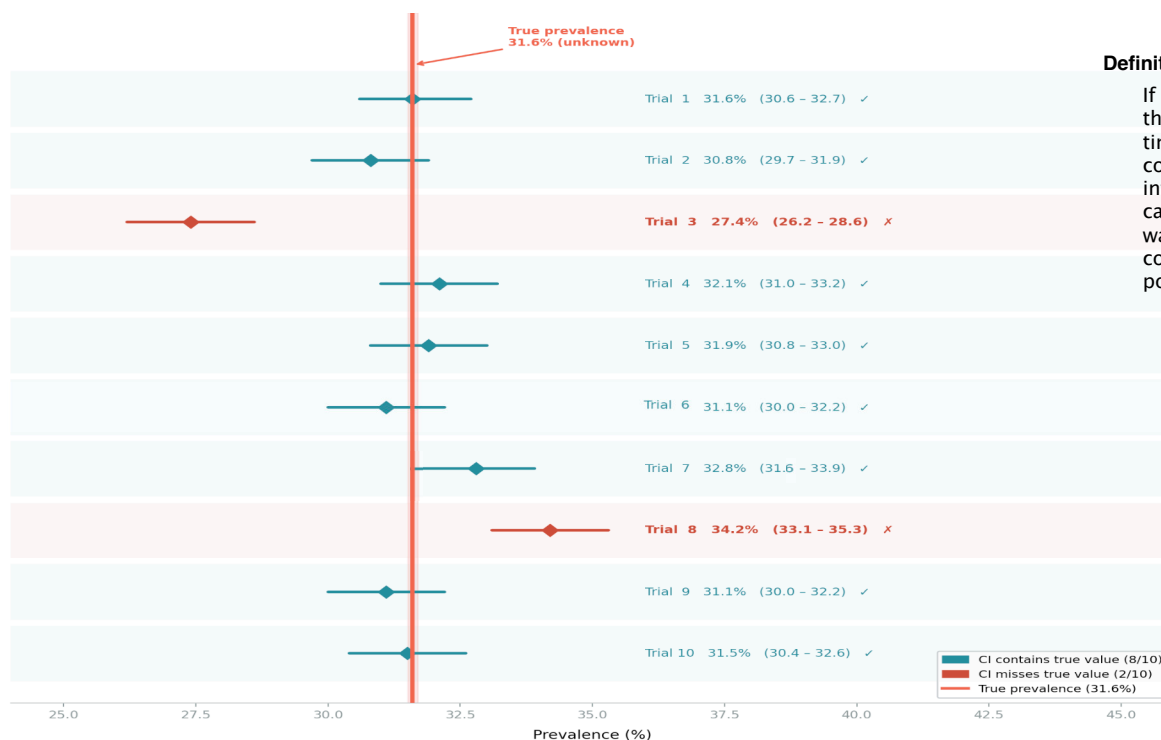


Figure. Top row: target population (true prevalence unknown) with repeated random sampling generating independent trial populations, each yielding a point estimate and 95% CI. Bottom: ten simulated CIs; Trials 3 and 8 (red) miss the true value — consistent with the ~5% expected miss rate.

Key Takeaways

1. The point estimate (♦) varies across repetitions — sampling variability is real.
2. CIs vary in position; with smaller samples they also vary in width.
3. ~5% of intervals miss the true value — that is what '95% confidence' means.
4. A single CI does not have a 95% probability of containing the truth; the truth is fixed — the interval is what is random.